

Cryptography: Transposition Ciphers

Unit III

Introduction to Transposition Ciphers

Definition

Transposition Ciphers are cryptographic techniques where the positions of characters in the plaintext are shifted according to a regular system.

Characteristics

Unlike substitution ciphers, transposition ciphers do not change the characters but rearrange them.

Common Examples

- **Rail Fence Cipher:** A simple zigzag pattern rearrangement.
- **Columnar Transposition Cipher:** Uses a keyword to determine column order.

Rail Fence Cipher

Description

- A form of transposition cipher that writes the message in a zigzag or “rail fence” pattern.
- After writing out the message in multiple “rails,” we read each rail sequentially to form the ciphertext.

Example

- Plaintext: HELLO WORLD
- Arrange in two rails:
 - Row 1: H L O O L
 - Row 2: E L W R D
- Ciphertext:
HLOOLELWRD

Columnar Transposition Cipher

Description

- In a columnar transposition cipher, the plaintext is written in rows, and the columns are rearranged according to a specific keyword.
- Each letter in the keyword corresponds to a column and determines the order in which columns are read.

Example

- Plaintext: ATTACK AT DAWN
- Keyword: CAB
- Arrange in columns:

C	A	B
A	T	T
T	A	C
K	A	T
D	A	W
N	-	-

- Ciphertext: TAAA TCTW ATKDN

Route Cipher

Description

- A Route Cipher writes the message in a grid, then reads it off in a set pattern.
- Common patterns include spirals, zigzags, and other routes across rows and columns.

Example

- Plaintext: DEFEND THE EAST WALL

- Write in a 3x5 grid:

D	E	F	E	N
D	T	H	E	A
S	T	W	A	L

- Ciphertext: DDS ETT FHW EEA NAL

Scytale Cipher

Description

- Ancient Greek cipher, using a strip of parchment wrapped around a rod (scytale).
- The message is only readable when wrapped around a rod of the same diameter.

Example

- Plaintext: HELLOWORLD
- Wrapped around a rod:
H E L L O
W O R L D
- Ciphertext: HWEOLRLLOD



Double Transposition Cipher

Description

- Applies two successive columnar transpositions with different keys.
- Creates additional complexity for decryption.

Example

- Plaintext: SECURITY
- First key: 4312
- Second key: 3142
- After applying both transpositions, Ciphertext: ECSUITYR

Double Transposition Cipher

Example

Step 1: First Transposition (Key: 4312)

Write the plaintext in columns according to the order of the key (4312). The number represents the order in which the columns will be rearranged.

Grid before the first transposition:

4	3	1	2
<i>S</i>	<i>E</i>	<i>C</i>	<i>U</i>
<i>R</i>	<i>I</i>	<i>T</i>	<i>Y</i>

Rearrange columns according to key 4312:

<i>C</i>	<i>U</i>	<i>E</i>	<i>S</i>
<i>T</i>	<i>Y</i>	<i>I</i>	<i>R</i>

After the first transposition, the text becomes: CTUYEISR

Double Transposition Cipher

Example

Step 2: Second Transposition (Key: 3142)

Now apply the second transposition using the key 3142. Write the result of the first transposition in columns:

3	1	4	2
<i>C</i>	<i>T</i>	<i>U</i>	<i>Y</i>
<i>E</i>	<i>I</i>	<i>S</i>	<i>R</i>

Rearrange columns according to key 3142:

<i>T</i>	<i>Y</i>	<i>C</i>	<i>U</i>
<i>I</i>	<i>R</i>	<i>E</i>	<i>S</i>

After the second transposition, the ciphertext becomes: TIYRCEUS

Myszkowski Transposition Cipher

Description:

- Similar to columnar transposition but uses repeated characters in the keyword.
- Columns with the same letter are read in sequence.

Example:

- Plaintext: HELLO WORLD
- Keyword: APPLE (A, P, P, L, E)
- Write in columns and read by columns based on the order of the keyword.

Disrupted Transposition Cipher

Description

- Divides text into sections that are scrambled or shifted.
- Adds additional complexity compared to regular transpositions.

Example

- Plaintext: SECURE MESSAGE
- Split into segments and scramble each.

Step 1: Split the plaintext into segments. Split the plaintext SECURE MESSAGE into 4-character segments:

SECURE MESSAGE → SEGMENTS: {SECU, RE M, ESSA, GE}

Step 2: Scramble each segment.

For each segment, rearrange the letters in a scrambled pattern:

- SECU becomes SCEU
- RE M becomes M RR
- ESSA becomes ESSA
- GE remains EG

Disrupted Transposition Cipher

Step 3: Combine the scrambled segments.

After scrambling the segments, combine them together:

Ciphertext: SCE UES GSM EGA

Grille Cipher

Description

- Uses a grid with cut-out sections to reveal portions of text.
- The grid is rotated to reveal additional characters.

Example

- Plaintext: HELLO WORLD
- Arrange plaintext in a grid:

<i>H</i>	<i>E</i>	<i>L</i>	<i>L</i>
<i>O</i>	<i>W</i>	<i>O</i>	<i>R</i>
<i>L</i>	<i>D</i>		

Grille Cipher - Step 1

Step 1: First Pass (No Rotation):

Read the letters visible in the cut-out sections (first and third columns):

Revealed: HLO

Step 2: Second Pass (90° Rotation):

After rotating the grid 90° clockwise, reveal the new letters:

L	O	L	E
W	O	W	R
D	R		

Revealed: *LOW*

Grille Cipher - Step 2

Step 3: Third Pass (180° Rotation):

Rotate the grid 180° and reveal more letters:

<i>L</i>	<i>W</i>	<i>D</i>	<i>L</i>
<i>O</i>	<i>O</i>	<i>R</i>	<i>H</i>
<i>L</i>	<i>R</i>		

Revealed: *LHD*

Grille Cipher - Step 3

Step 4: Fourth Pass (270° Rotation):

Rotate the grid 270° and reveal the final set of characters:

<i>L</i>	<i>E</i>	<i>D</i>	<i>L</i>
<i>O</i>	<i>R</i>	<i>W</i>	<i>H</i>
<i>L</i>	<i>D</i>		

Revealed: *LWR*

Final Ciphertext: HLO LOW LHD LWR